Questions by topic: Manipulation of food species & agricultural energetics - Set 1 (80 marks)

Answer all questions in the spaces provided.

0 1 . 1 Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0 1 . 2 Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

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0 2 Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

0 2 . 1 Explain the main advantage of GM over conventional selective breeding.

[2 marks]

0 2 . 2 For each crop in Table 1, explain one potential benefit.

[3 marks]

Bt crops

Roundup Ready soya

Golden rice

0 2 . 3 Explain **four** concerns about GM crops.

[4 marks]

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0 2 . 4 Give **one** reason why GM crops have been slow to enter cultivation in Europe.

[1 mark]

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0 3 Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 3 . 1 Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 3 . 2 Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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0 3 . 3 Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

0 3 . 4 Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

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0 4 Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

0 4 . 1 Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

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0 4 . 2 Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

0 4 . 3 Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

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0 5 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 5 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 5 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

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0 6 Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 6 . 1 Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

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Answer [ ] %

0 6 . 2 Explain why so little of the solar energy is stored in the crop.

[3 marks]

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0 6 . 3 Explain why few food chains have more than four trophic levels.

[2 marks]

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0 6 . 4 Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

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0 6 . 5 Explain how pigs can increase the efficiency of a food production system.

[2 marks]

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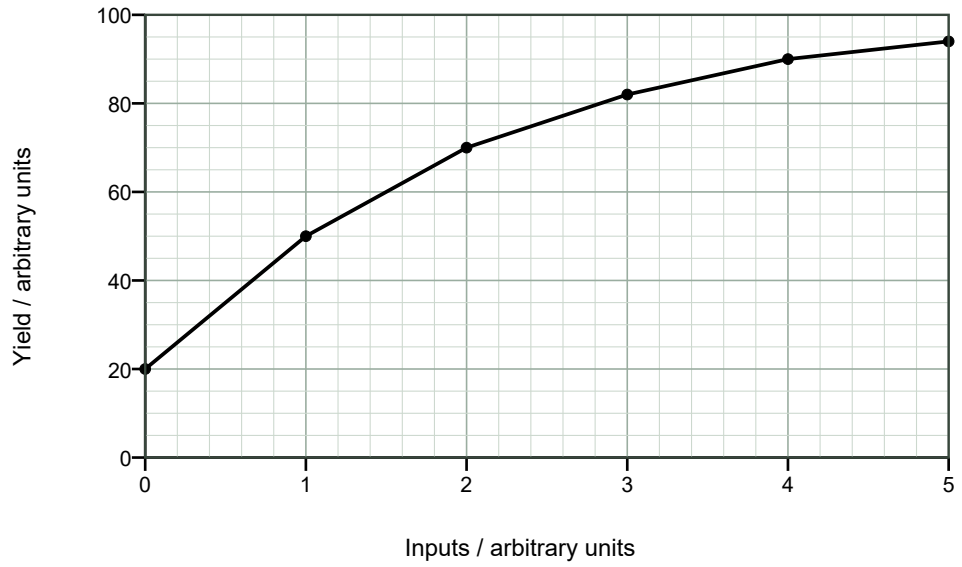
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0 7 Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



0 7 . 1 Describe the relationship shown in Figure 1.

[2 marks]

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0 7 . 2 Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

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Answer [ ] units

0 7 . 3 Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

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0 7 . 4 Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

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0 7 . 5 Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

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0 8 Table 1 shows the energy inputs and food energy outputs of three farming systems, in GJ ha^{-1} yr^{-1}.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Subsistence maize, hand cultivated	0.6	9.0
Intensive wheat, UK	18.0	54.0
Intensive beef, feedlot	40.0	12.0

0 8 . 1 Calculate the energy ratio (output / input) for each system in Table 1.

[3 marks]

Answer

0 8 . 2 Explain what is meant by an **energy subsidy**, and use your answers to explain why the intensive systems have much lower energy ratios.

[3 marks]

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0 8 . 3 Explain **three** ways a farmer can reduce the energy losses in a food production chain.

[3 marks]

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0 8 . 4 Explain **two** advantages of genetic modification over conventional selective breeding.

[2 marks]

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0 8 . 5 Explain **four** concerns about the use of GM crops.

[4 marks]

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END OF QUESTIONS